

Sort Green Coffee Beans by Grade SOP

One episode sorts one scoop of green coffee beans. It runs from a bare tray, a bin holding eight beans, an empty cup and two empty tins, to the eight beans split three ways: the two defect beans in the cup, the three big beans in the big tin, the three small beans in the small tin, the big tin weighed on the scale, four boxes marked on the log card and every tool back on its spot.

The order never changes: **spread the scoop on the tray, pick the defect beans into the cup, sort the rest by size into the tins, weigh the big tin, mark the log.** Nothing is sorted by size until every defect bean is off the tray, and nothing is weighed until the tray is bare.

One arm does all of it. The right gripper scoops, spreads, picks every bean, carries the tin to the scale and marks the log card. The left arm stays at home with its gripper open for the whole episode. It never enters the work area, never touches an object and never steadies one. Nothing in this task has to be held, because nothing moves on its own: the tray, the cup and both tins sit in holders that grip the table, so the one arm can pick out of them and drop into them with no second hand. The right arm is the working arm because the rig is set up right handed, so the dominant hand drives it. A left handed collection requires a separately approved mirrored variant. Do not swap arms during an episode.

Every spot sits on the right arm's side or on the midline, inside its comfortable reach, so the arm never has to stretch or fold in on itself. Only one object is held at a time, no grasp is changed while an object is off the table, and no bean is ever carried over a container it is not going into. Each bean leaves the tray by the edge nearest the container it belongs in: defects off the left edge, big beans off the right edge, small beans off the near edge.

Each tool goes back to its own spot before the next step starts. The counts are fixed: one level scoop, eight beans on the tray, two defect beans, three big beans, three small beans, one weighing and nine marks. There is no second scoop and no second weighing. A bean that lands off the tray or off the table is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 90 mm.
- **cell_type:** single_arm
- **parked_side:** left

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the bean bin, the scoop spot, the sorting tray, the defect cup, both tin spots, the scale, the log card and the marker holder.
3. The tray surface is visible from above, so every bean on it can be counted, and the scale display and the four boxes on the log card both stay in frame the whole time.
4. The right arm is at home with its gripper open. The left arm is at home with its gripper open and stays there.
5. The left arm is parked clear of the work area and of the camera's view of the tray, the scale display and the log card. If it blocks any of them, move it further out and check the frame again.
6. The tabletop is dry and clear of anything but the items in the Materials checklist.
7. The right arm reaches every spot on the table, the bean bin, the scoop spot, the tray, the defect cup, both tins, the scale, the log card and the marker holder, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
8. The right arm reaches the far edge of the tray and the near edge of the tray, so it can pick a bean from anywhere on the tray surface.
9. Nothing needs the far side or the left side of the table. The paths from the bean bin to the tray, from the tray to the cup, from the tray to each tin, from the big tin to the scale and from the marker holder to the log card are unobstructed. Carry one bean along each path by hand during setup: it must clear the rim of every container it passes.

Materials checklist

1. One sorting tray sits at the middle of the table, straddling the midline, bare and dry. It is flat bottomed with a low lip, and wide enough to hold eight beans spread out in one layer with a clear gap between them.
2. The tray sits in a holder that grips the table. Push it by hand during setup: if it slides, turns or lifts, weight the holder or replace it. Nothing pins the tray during the episode.
3. One bean bin stands open at the bean spot at the back, holding exactly eight green coffee beans and nothing else. It is low enough that the scoop reaches the beans without the wrist passing below the bin rim.
4. The eight beans are the same eight in every episode: two defect beans, three big beans and three small beans.
5. The two defect beans are plainly bad from above: each one is broken clean in half, so a reviewer can tell it apart from a whole bean in a single frame. A bean that is only faintly off colour is not used.

6. The three big beans and the three small beans are two clearly different sizes, at least two screen sizes apart, so the choice is plain to see. Beans that are close in size are replaced.
7. One size card lies flat beside the tray on the right, with one big bean and one small bean glued to it, labelled BIG and SMALL. It is never picked up and never moved.
8. One defect cup stands in its holder just left of the tray, empty and dry, with a mouth wide enough to drop a bean into from above.
9. Two tins stand in their rings right of the tray, empty and dry: the big tin nearer the tray and the small tin beyond it. The two are clearly different in size or colour, and the same tin is the big tin in every episode.
10. Each tin and the cup is low enough and wide enough that a bean can be dropped in from just above the rim, and each one sits in its ring so it cannot tip or slide when a bean lands in it.
11. One scoop lies flat on the scoop spot, back right, bowl down, handle toward the right arm. It is wide and shallow, big enough to lift all eight beans in one pass, so a quarter turn empties it.
12. One scale stands at the front right, directly in front of the big tin spot, switched on, level, and reading zero with nothing on it. Its pan is big enough for the big tin to stand fully on it, and its display faces the environment camera.
13. The scale is zeroed by hand during setup. Nothing on this station presses a scale button during the episode.
14. One log card lies flat at the front centre on the midline. It is a wipe clean card with one row of four boxes on it, labelled DEFECT, BIG, SMALL and WEIGHED, each box big enough to hold three strokes with room to spare, and it starts the episode blank.
15. The log card grips the table. Push it by hand during setup: if it slides, tape it down or put it in a holder.
16. One dry erase marker stands in its holder on the log card's right edge, cap off, with the tip toward the right arm and enough ink to draw nine marks.

Workspace layout

Four rows: the beans and the scoop at the back, the tray in the middle with the cup on its left and the tins on its right, the scale in front of the tins, and the log card at the front. Everything travels forward and outward, off the tray and into a container.

Back row, left to right:

1. Bean spot, back centre on the midline, the bean bin standing open on it.
2. Scoop spot, back right, on the right arm's side.

Middle row, left to right:

3. Defect cup spot, left of the tray, the cup standing in its holder. The defect beans leave the tray by its left edge.
4. Sorting tray, centre, straddling the midline. It is bare at the start of the episode and bare again at the end.
5. Big tin spot, right of the tray, the tin standing in its ring. The big beans leave the tray by its right edge.
6. Small tin spot, right of the big tin spot, the tin standing in its ring, with the size card flat on the table between the two tins.

Third row, left to right:

7. Small bean drop path, straight forward off the tray's near edge, then right along the front of the tray to the small tin. It passes in front of the tins, never over them.
8. Scale, front right, directly in front of the big tin spot, with its display facing the camera.

Front row:

9. Log card, front centre on the midline, its four boxes in one row facing the camera.
10. Marker holder, on the log card's right edge, on the right arm's side.

A clear gap of at least a gripper's width sits between the tray and the cup, between the tray and the big tin, and between the two tins, so a bean drops into each one straight from above without the gripper touching the next container. No carry passes over a container the bean is not going into, and the arm never travels to the far side or the left side of the table.

Vocabulary

- **Arm assignment:** the right gripper does everything: the scoop, the spread, every bean pick, the weighing and the marks. The left arm is parked and does nothing at all.
- **Parked arm:** the left arm, at home with its gripper open, still for the whole episode. It never reaches in, never holds anything, and never blocks the camera.
- **Tool home:** the spot a tool returns to and lies flat or stands in before the next step starts: the scoop spot, the marker holder. A tool is never left on the tray, in a tin, on the scale or in the gripper between steps.
- **Release point:** the gripper opens just above the tray when beans are being spread, just above the rim of the cup or a tin when a bean is being dropped, over the scale pan when the tin is being set down, or over a tool home. Never over the wrong container and never over open tabletop.
- **Stable placement:** the object stands still after release, flat on its surface, and does not rock, tip, slide or lean on anything.
- **Green bean:** one whole raw coffee bean. There are eight on the tray at the start of Step 2 and none at the end of Step 3.

- **Defect bean:** a bean broken clean in half. There are two in every episode and they go to the cup and nowhere else.
- **Sound bean:** a whole unbroken bean. There are six in every episode and each one is either a big bean or a small bean.
- **Big bean and small bean:** the two sizes the sound beans come in, clearly apart, matched against the size card if there is any doubt. Three of each in every episode.
- **Size card:** the card beside the tray with one big bean and one small bean glued to it. It is looked at, never touched.
- **Single layer:** the beans lying on the tray floor, none on top of another and none touching, so every bean can be seen and picked on its own.
- **Level scoop:** a scoopful with no bean standing above the scoop rim. A heaped scoop is not level.
- **One bean per pick:** the gripper closes on one bean, carries it, drops it, and comes back empty before it closes on the next one. Two beans in one pick is not a pick.
- **Bare tray:** no bean anywhere on the tray surface, and nothing on it but the tray itself.
- **Settled reading:** the number on the scale display has stopped changing and holds still for 2 seconds with nothing touching the tin.
- **Tally stroke:** one short straight line drawn in a box on the log card, one for each bean counted, drawn left to right along the box.
- **Marked box:** a box on the log card holding its marks, all of them inside its lines and none touching the next box.
- **Spill:** a bean outside the tray and outside every container. It is left where it falls and cleared in the reset, never on camera.
- **Quarter turn:** the most any object is tilted in the gripper. Scooping, spreading and dropping all stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Spread one scoop on the tray

Goal: all eight beans lying on the tray in one layer, none on top of another, the bean bin empty, and the scoop back on its spot.

The tray and the bin both grip the table on their own, so nothing holds either of them while the scoop works. Neither is ever lifted.

1.1 Take the scoop

- With the **right gripper**, close on the scoop handle end, lift it straight up off the scoop spot, and carry it to the bean bin.

Check: the scoop hangs level in the gripper and does not slip. If it hangs tilted, lay it back on its spot and take it again. Make no more than two corrective takes. If it still will not hang level, end the episode and correct the scoop during reset.

1.2 Scoop the beans

- Lower the scoop into the bin bowl down, and push it forward through the beans until all eight are in it.
- Draw the scoop up the inside wall of the bin as you lift, so any bean standing proud falls back in, until no bean stands above the rim.
- Lift the scoop clear of the bin, keep it level, and carry it forward to the tray.

One scoop, and one only. If a bean is left in the bin, end the episode and let review flag it. There is no second scoop to fetch it.

Check: the bin is empty and the scoop holds eight beans, level, with none above the rim.

1.3 Spread the beans on the tray

- Bring the scoop over the tray, low to it and inside its lip, and stop before you tilt.
- Tilt the scoop a quarter turn and let the beans run out onto the tray floor, working the scoop slowly from the tray's left side to its right side as they fall, so they land spread out and not in one pile.
- Bring the scoop back to level before it leaves the tray.
- If any bean has landed on another, pick it up with the gripper and set it down on a bare part of the tray. Beans are moved one at a time and never pushed or raked.

The beans have to lie in one layer, because a bean under another bean cannot be seen, graded or counted.

Check: eight beans lie on the tray, all on the tray floor, none on top of another, none touching, and none over the lip. If two beans touch, move one of them once. Make no more than two corrective moves.

1.4 Park the scoop

- With the **right gripper**, carry the scoop back to the scoop spot, lay it flat bowl down, and release.

Check: the scoop lies fully on its spot, not on the tray, not in the bin, and not over the table edge.

Step 2: Pick the defect beans into the cup

Goal: both defect beans in the cup, six sound beans still on the tray, and nothing else moved.

The defects come off first, so the size sort in Step 3 only ever looks at sound beans. A broken bean is not big or small: it has no grade at all.

- Look over the tray and find a bean that is broken in half.
- With the **right gripper**, close on that bean on its own, lift it straight up clear of the tray, and carry it off the tray's **left edge** to the defect cup.
- Bring it over the cup mouth, low to the rim, stop, and open the gripper so it drops straight in.
- Come back empty and repeat for the second defect bean.
- One bean per pick. The gripper never closes on two beans, and never carries a defect bean over a tin.

Check: two beans lie in the cup, six whole beans lie on the tray, and none of the six is broken. If a bean has been dropped outside the cup, leave it where it fell and let review flag the episode. If a sound bean has gone into the cup, end the episode and reset.

Step 3: Sort the sound beans by size into the tins

Goal: three big beans in the big tin, three small beans in the small tin, and the tray bare.

Both tins stand in their rings, so a bean can be dropped into either one without anything holding it. Neither tin is lifted in this step.

3.1 Big beans to the big tin

- Look over the six beans and find the big ones, matching them against the size card.
- With the **right gripper**, close on one big bean on its own, lift it straight up clear of the tray, and carry it off the tray's **right edge** to the big tin.
- Bring it over the tin mouth, low to the rim, stop, and open the gripper so it drops straight in.
- Come back empty and repeat until all three big beans are in the big tin.

Check: three beans lie in the big tin and three beans are left on the tray. If a big bean has gone into the small tin, end the episode and reset. Beans are never tipped back out of a tin.

3.2 Small beans to the small tin

- With the **right gripper**, close on one small bean on its own, lift it straight up clear of the tray, and carry it off the tray's **near edge**, then right along the front of the tray to the small tin.
- Bring it over the tin mouth, low to the rim, stop, and open the gripper so it drops straight in.
- Come back empty and repeat until all three small beans are in the small tin.
- The small beans go round the front of the big tin, never over it.

Check: the tray is bare, three beans lie in the small tin, three in the big tin and two in the cup. If a bean is still on the tray, look at it again against the size card and place it. Make no more than one correction. If the counts do not come to eight, end the episode and let review flag it.

Step 4: Weigh the big tin

Goal: the big tin standing alone on the scale, a settled reading in frame, and the tin back in its ring.

The scale is already reading zero and nobody touches its buttons. The tin travels straight forward from its ring onto the pan, so it passes over nothing on the way.

4.1 Set the tin on the scale

- **Check:** the scale reads zero with nothing on its pan. If it does not, end the episode and zero it during reset.
- With the **right gripper**, close on the big tin's rim, square to the tin so it hangs upright and level. The rim is the grasp: the gripper cannot span a tin wall securely.
- Lift it straight up out of its ring, clear of the table, and carry it forward level, without tilting or swinging it, onto the middle of the scale pan.
- Lower it until it takes its own weight, then open the gripper and lift clear.

The beans stay in the tin. Nothing is ever poured out to weigh it.

4.2 Let the reading settle

- Take the gripper right away from the tin and the pan and hold still.
- Wait until the number on the display stops changing and holds still for 2 seconds. That is the reading.
- The camera records the reading. The number is not written down anywhere by the arm.

Check: the tin stands fully on the pan, upright, touching nothing but the pan, and the display is readable in frame. If the tin overhangs the pan or leans on anything, lift it clear and set it down again, once.

4.3 Return the tin to its ring

- With the **right gripper**, close on the tin's rim square, lift it straight up off the pan, carry it back level to the big tin spot, and lower it into its ring.
- Release once it stands on its own, then lift clear.

Check: the tin sits flat in its ring without rocking, still holding its three beans, and the scale reads zero again with a bare pan. If the tin rocks, lift it clear and seat it again, once.

One weighing, and one only. The tin does not go back on the scale to check the number.

Step 5: Mark the counts on the log card

Goal: two strokes in the DEFECT box, three in the BIG box, three in the SMALL box, one X in the WEIGHED box, and the marker back in its holder.

The log card grips the table, so nothing holds it while it is marked. The marks are made left to right along the row, in the order the beans were sorted.

5.1 Take the marker

- With the **right gripper**, close on the marker barrel, lift it straight up out of its holder, and bring the tip over the log card.

Check: the marker hangs upright with its tip down and does not slip in the gripper.

5.2 Mark the three count boxes

- Draw **two** short strokes in the DEFECT box, side by side, left to right, both fully inside the box lines.
- Draw **three** strokes the same way in the BIG box.
- Draw **three** strokes the same way in the SMALL box.
- One stroke stands for one bean. Lift the tip clear of the card between strokes so they do not run together.

The counts are read off the containers, not from memory: two in the cup, three in the big tin, three in the small tin.

Check: the three boxes hold two, three and three strokes, every stroke is inside its own box, and no stroke crosses a box line. A wrong count is not rubbed out and not drawn over. Leave it and let review flag the episode.

5.3 Mark the weighed box

- Draw one X in the WEIGHED box, fully inside its lines.

The X means the big tin was weighed. It goes on only after the tin is back in its ring.

5.4 Park the marker

- With the **right gripper**, carry the marker back to its holder, lower it in until it stands on its own, release, and lift clear.

Check: the marker stands in its holder, tip down, and no marker is left lying on the card or the table.

Step 6: End the episode

- Confirm the tray: bare, with no bean anywhere on it.
- Confirm the containers: two beans in the cup, three in the big tin, three in the small tin, each container upright in its holder or ring.
- Confirm the scale: bare pan, reading zero.
- Confirm the log card: two, three, three and one X, each mark inside its own box.
- Confirm the tools: the scoop flat on its spot, the marker upright in its holder, the size card unmoved.
- Confirm the left arm: still at home, gripper open, and it never moved.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the right arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Tip the beans out of the cup and both tins and back into the bean bin, and stand the cup and the tins back in their holders, empty and dry.
2. Count the beans back into the bin: exactly eight, two broken and six whole, three big and three small. Replace any bean that has cracked, chipped or changed size so the eight are the same eight next time.
3. Pick up any bean that fell off the tray or off the table and put it back in the bin. Replace it if it is damaged.
4. Wipe the log card clean and dry it, so all four boxes start blank.
5. Wipe any dust or chaff off the tray, the scale pan and the tabletop, and dry them.
6. Lay the scoop flat on its spot bowl down with the handle toward the right arm, and stand the marker in its holder with the cap off. Replace a marker that is running dry.
7. Check the scale reads zero with a bare pan, and zero it by hand if it does not.
8. Confirm the tray, the cup, both tins and the log card all still grip the table, and that no tin or cup is dented so it will not stand upright.
9. Confirm the size card still lies flat beside the tray with both its beans attached.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a sound bean is sorted by size while a defect bean is still on the tray, the tin goes onto the scale before the tray is bare, a mark goes on the log card before the tin is back in its ring, the X goes in the WEIGHED box before the weighing, or a step is skipped.

- **SOP rule broken:** Steps 1 to 5, spread, pick the defects, sort by size, weigh, mark, in that order, with every defect off the tray before any size sort starts.
- **Coaching note:** spread it, pick the bad ones, sort the rest, weigh it, mark it. The defects come off first, every time.

Violation: Parked arm moved, or the wrong arm used

- **Visible cue:** the left arm leaves home, reaches into the work area, closes its gripper, touches an object, steadies a container, or drifts in front of the tray, the scale display or the log card. Any object taken or carried by the left gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 6, this is a single arm task: the right gripper does all of it and the left arm stays at home with its gripper open for the whole episode.
- **Coaching note:** one arm only. If the left arm has moved, the episode is a bimanual episode wearing the wrong label.

Violation: Container held or steadied

- **Visible cue:** the gripper presses on the tray, the cup, a tin, the scale or the log card to keep it still while working, instead of letting the holder do it.
- **SOP rule broken:** Steps 1 to 5, every container grips the table or stands in its holder, and nothing on this station is held.
- **Coaching note:** if something needs holding, the holder is wrong. Fix the station, do not add a hand.

Violation: Scoop heaped, or emptied from the wrong place

- **Visible cue:** a bean stands above the scoop rim as the gripper lifts the scoop out of the bin, with no draw up the inside wall to level it; the scoop tilts before it is over the tray, is held outside the tray lip, or is held high so the beans fall a visible distance; or the scoop leaves the tray before it is back to level.
- **SOP rule broken:** Steps 1.2 and 1.3, level the scoop against the inside wall of the bin, bring it inside the tray lip and low to it before tilting, and bring it back to level before it leaves the tray.
- **Coaching note:** ride the inside wall on the way up. Finish the carry, stop, then tilt. Level first, then leave.

Violation: Beans not in a single layer

- **Visible cue:** a bean rests on top of another bean, beans sit in a pile or a heap, or two beans touch, when Step 2 begins.
- **SOP rule broken:** Step 1.3, the eight beans lie on the tray floor in one layer, none on top of another and none touching.
- **Coaching note:** a bean you cannot see is a bean you cannot grade. Spread it out before you start picking.

Violation: Beans pushed, raked or swept

- **Visible cue:** the gripper pushes, drags, rolls, rakes or sweeps beans across the tray, or tips the tray to move them, instead of lifting each one.
- **SOP rule broken:** Steps 1.3, 2 and 3, a bean is moved by being picked up and set down, and by nothing else.
- **Coaching note:** pick it up, do not push it. A pushed bean tells the model nothing about grasping.

Violation: More than one bean per pick

- **Visible cue:** the gripper closes on two or more beans at once, carries two beans in one trip, or drops two beans into a container in one release.
- **SOP rule broken:** Steps 2 and 3, one bean per pick: close on one bean, carry it, drop it, and come back empty.
- **Coaching note:** one bean, one trip. A double pick loses the count and the grasp.

Violation: Bean put in the wrong place

- **Visible cue:** a defect bean goes into a tin, a sound bean goes into the cup, a big bean goes into the small tin, a small bean goes into the big tin, or a bean is dropped back onto the tray after Step 3 has started.
- **SOP rule broken:** Steps 2 and 3, defects to the cup, big beans to the big tin, small beans to the small tin.
- **Coaching note:** look at the bean, then look at the size card, then move. The grade is decided before the gripper closes.

Violation: Bean tipped back out to correct a sort

- **Visible cue:** a bean is lifted back out of the cup or a tin, or a container is tipped, to move a bean that was put in the wrong place.
- **SOP rule broken:** Step 3.1, beans are never tipped back out of a container, and a wrong sort is tagged, not fixed.
- **Coaching note:** once it is in, it stays in. A correction pick makes the episode unlearnable.

Violation: Tray not bare at the end of the sort

- **Visible cue:** a bean is still on the tray when the tin goes onto the scale, or the eight beans do not come to two in the cup, three in the big tin and three in the small tin.
- **SOP rule broken:** Step 3.2, the tray is bare and the counts come to eight before Step 4 starts.
- **Coaching note:** count the tray before you leave it. Eight went on, eight come off.

Violation: Bean released before it is over its container

- **Visible cue:** the gripper opens while the bean is still over the tray, over open tabletop, or over the wrong container, or it opens high so the bean bounces out.
- **SOP rule broken:** Steps 2 and 3, the gripper opens low over the mouth of the container the bean belongs in.
- **Coaching note:** carry it all the way, low to the rim, then open. A bean dropped early is a bean on the floor.

Violation: A bean carried over the wrong container

- **Visible cue:** a defect bean passes over a tin, a big bean passes over the small tin or the cup, or a small bean passes over the big tin on its way to its own tin.
- **SOP rule broken:** Steps 2, 3.1 and 3.2, each bean leaves the tray by the edge nearest its container: defects left, big beans right, small beans forward and round the front.
- **Coaching note:** route around, not across. A bean dropped over the wrong tin lands in the wrong tin.

Violation: Weighing done wrong

- **Visible cue:** the tin is set down off the middle of the pan or overhanging it, the gripper stays on the tin or touches the pan while the number settles, the tin leans on anything, or the reading is called before the number has held still for 2 seconds.
- **SOP rule broken:** Steps 4.1 and 4.2, the tin stands alone in the middle of the pan, touching nothing but the pan, and the reading is the number that has stopped changing.
- **Coaching note:** set it down, let go, back off, then read. A hand on the tin is a hand on the scale.

Violation: Wrong tin weighed, or the tin emptied to weigh it

- **Visible cue:** the small tin goes onto the scale instead of the big tin, both tins go on, or the beans are tipped out of the tin and weighed loose on the pan.
- **SOP rule broken:** Step 4, the big tin is weighed once, with its three beans still in it.
- **Coaching note:** the big tin, beans and all. It is weighed as it stands.

Violation: Second weighing, or the scale touched

- **Visible cue:** the tin goes back onto the scale after it has been returned to its ring, or the gripper presses a scale button, lifts the pan, or nudges the scale at any point.
- **SOP rule broken:** Steps 4.2 and 4.3, one weighing and one only, and the scale is zeroed by hand during setup, never on camera.
- **Coaching note:** one reading. If it looked wrong, tag it and let review decide.

Violation: Marks wrong

- **Visible cue:** the strokes do not match the counts, marks go in the wrong box, a stroke crosses or sits outside a box line, strokes run together so they cannot be counted, a mark is rubbed out or drawn over, or the WEIGHED box is left blank.
- **SOP rule broken:** Steps 5.2 and 5.3, two strokes in DEFECT, three in BIG, three in SMALL, one X in WEIGHED, each mark fully inside its own box.
- **Coaching note:** read the counts off the containers, then mark. A wrong count is tagged, never rubbed out.

Violation: Spill cleaned up

- **Visible cue:** after a bean lands on the table or on the floor, the gripper is used to sweep, push, or pick it up.
- **SOP rule broken:** Steps 1 to 5, a spill is left where it falls and is cleared in the reset.
- **Coaching note:** cleanup motions are not part of the task. The spill is data, the cleanup is noise.

Violation: Bad pick, carry, or release

- **Visible cue:** an object is dragged along the table with no daylight under it, the gripper shifts or seats its hold again while the object is off the table, a tin hangs tilted in the gripper because the rim grasp was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 5, one object at a time, lifted straight up clear of the surface before any horizontal motion, a tin gripped square on its rim, and tilted no more than a quarter turn.
- **Coaching note:** set it down to change the grip, and lift before the carry. A tin that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a tin, the cup, the tray, the scoop, the marker or the log card is shoved, nudged, or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 5, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the scoop is left in the bin, on the tray or in a tin, the marker is left lying on the log card or the table, the gripper opens while the marker is still above its holder, or a tool is still in the gripper when the next step begins.
- **SOP rule broken:** Steps 1.4 and 5.4, each tool returns to its home and lies flat or stands in it before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Size card touched or moved

- **Visible cue:** the gripper picks up, slides or knocks the size card, or lays a bean on it to compare.
- **SOP rule broken:** Materials checklist and Step 3.1, the size card is looked at and never touched.
- **Coaching note:** compare with the eyes. The card is a reference, not a work surface.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried object strikes the tray, the cup, a tin, the bin, the scale or the log card and knocks, tips, slides, or displaces it; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 5, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. The containers stand close together, so the route matters more than the reach.

Violation: Manipulation outside the camera frame

- **Visible cue:** a scoop, a spread, a bean pick, a drop, the weighing or a mark happens partly or wholly outside the environment camera frame, or the parked left arm blocks the camera's view of the tray, the scale display or the log card.
- **SOP rule broken:** Steps 1 to 5, every manipulation happens inside the camera frame and nothing blocks it.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and park the left arm further out.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a bean still on the tray, a container in the wrong place, the tin still on the scale, a box unmarked, a tool out of its home, an arm short of home, or a gripper not fully open.

- **SOP rule broken:** Step 6, confirm the tray, the containers, the scale, the log card and the tools, then return the right arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective tray: one that slides, turns or lifts in its holder under a correct pick, or that will not lie flat.
- Defective cup or tin: one that will not stand upright in its holder, whose rim the gripper cannot close on squarely, or that is dented or cracked.
- Defective scoop: one too small to lift all eight beans in one pass, or that needs more than a quarter turn to empty.
- Defective scale: one that will not hold zero, whose number never settles, whose pan is too small for the tin, or whose display is unreadable in frame.
- Dry marker, or a log card that will not wipe clean so a box starts the episode already marked.
- Beans too close in size to tell apart, or a defect bean that is not plainly broken. Replace the beans before the next episode.
- Bean bin holding fewer or more than eight beans at episode start, or the wrong mix of them. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take the scoop from its spot
2. Scoop the beans from the bin
3. Spread the beans on the tray
4. Return the scoop to its spot
5. Pick one defect bean from the tray
6. Drop one defect bean into the cup
7. Pick one big bean from the tray

8. Drop one big bean into the big tin
9. Pick one small bean from the tray
10. Drop one small bean into the small tin
11. Lift the big tin from its ring
12. Set the big tin on the scale
13. Return the big tin to its ring
14. Take the marker from its holder
15. Draw one stroke in a count box
16. Draw one X in the weighed box
17. Return the marker to its holder
18. Return the arm home and end the episode